Programming(7)

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Learning Objectives

- Understand and write nested if statements
- Recognize when nested conditions are needed
- Build multi-level decision structures
- Import and use Python modules
- Create a simple custom module
- Organize code into separate files
- Understand modular programming



Nested Conditions



Why Nested Conditions?

We use nested conditions when:

A decision depends on another decision

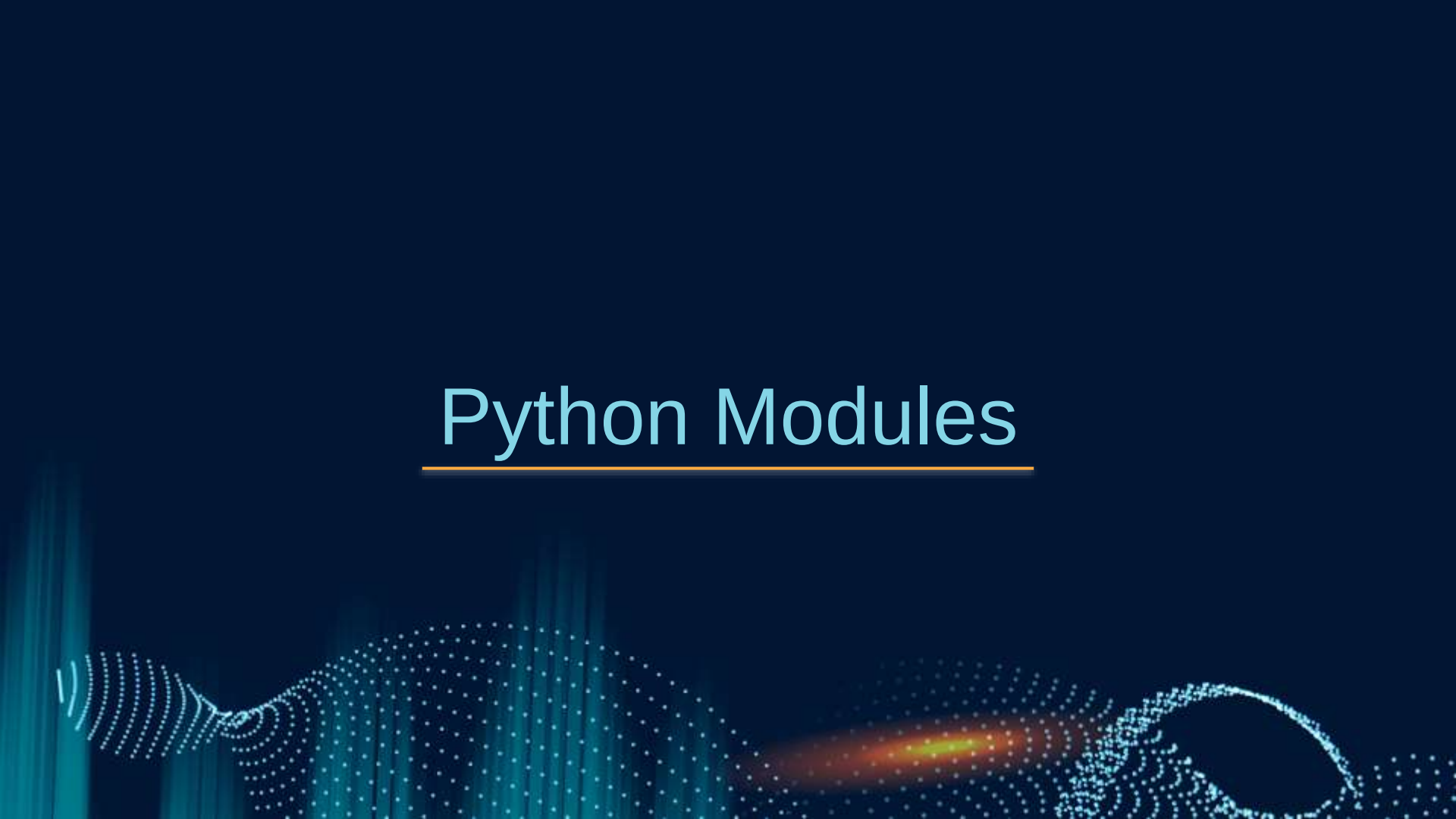
Multi-level checking is needed

Input validation requires steps

The problem includes “first check X, then inside it check Y”



Python Modules



What are Modules?

A module is a file containing Python code that we can import and reuse.

Reasons for modules:

Code organization

Reusability

Clean and scalable programs

Separation of logic (login module, item module, main program)



Python Modules

Using math module

index.py

```python
#Basic import:
import math


#Using only part of a module:
from math import sqrt


#Import with alias:
#import math as m

print(math.sqrt(16))      # 4.0
print(math.pi)            # 3.14...
print(math.factorial(5)) # 120

```



Thank You

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